

Resolution Geometry as an Observable Orbit–Cone Structure

Quotients, Automorphism Groupoids, Saturation, Transport, and Conformal Reconstruction

Matěj Rada

MatejRada@email.cz

Licensed under CC BY-NC-ND 4.0

August 24, 2026

Abstract

Resolution Geometry (RG) has two apparently opposite descriptions. The top-down description begins with an observation law and collapses observationally indistinguishable latent realizations. The bottom-up description begins with an admissible automorphism groupoid and constructs its orbit space and invariant coordinates. We show how these descriptions agree, and precisely when they do not. Gauge soundness gives a canonical surjection

$$X_{\text{lat}}/\mathcal{A} \longrightarrow Q_{\text{obs}},$$

but equality requires the additional gauge-completeness condition that declared automorphisms generate every observation fibre. The canonical comparison object is the kernel groupoid

$$\mathcal{K}_{\text{obs}} = X_{\text{lat}} \times_{Q_{\text{obs}}} X_{\text{lat}} \rightrightarrows X_{\text{lat}}.$$

Saturation, gauge completeness, and latent reconstruction are then separated as three different claims. On the regular stratum the common differential object is the observable horizontal tangent structure

$$T_{[x]}Q_{\text{obs}} \cong T_x X_{\text{lat}} / \ker D\pi_{\text{obs}},$$

while at isotropy and rank jumps it is replaced by slice and tangent-cone data. A latent transport descends uniquely to W_{obs} exactly when it is observationally projectable, and an admissible latent cone descends exactly when its projection is constant along complete observation fibres. A regular quadratic descended cone determines a Lorentzian conformal class $[g]$, whereas nonquadratic and singular cones require wider characteristic or stratified geometry. An integrable operational scale field is still required to select a metric representative. A deterministic hostile battery, seed 20260824, passed all 26 registered tests and forced the theorem ladder stated here. The result is a single stratified geometry in which the quotient, groupoid, saturated fibre, observable transport, and cone are complementary layers rather than competing foundations.

Contents

1	The apparent duality	2
2	Observation quotient and kernel groupoid	2
3	Saturation is not gauge completeness	3
4	The mutual fibre and horizontal observable structure	4
5	Descent of observable transport	5
6	Descent of the admissible cone	5

7	From the observable cone to conformal geometry	6
8	Scale selection and strict transport	7
9	The Observable Orbit–Cone Theorem	7
10	Why this is the full geometry of RG	8
11	Attack battery and reproducibility	9
12	Non-claims	10
13	Conclusion	10

1 The apparent duality

The RG corpus establishes three facts. First, exact observable data (J, Σ, g_O) canonically determine the resolved sector $R = \text{Im } J$, the null sector $N = \ker J^* = R^{\perp_{g_O}}$, and the full covariance block geometry; admissible isometries act equivariantly. Second, the natural fibre is not a manifold but a stratified orbit space containing spectral chambers, Grassmannian angle data, positive-semidefinite cones, rank walls, and Schur-admissibility boundaries. Third, generic second-order saturation needs both magnitude and frame-orientation invariants; canonical correlations alone are incomplete.

These results invite two constructions:

$$\begin{aligned} \text{top-down:} \quad & X_{\text{lat}} \xrightarrow{\pi_{\text{obs}}} Q_{\text{obs}} = X_{\text{lat}} / \sim_{\text{obs}}, \\ \text{bottom-up:} \quad & \mathcal{A} \rightrightarrows X_{\text{lat}} \longrightarrow X_{\text{lat}} / \mathcal{A}. \end{aligned}$$

The first is defined by what observations fail to distinguish. The second is defined by transformations known to preserve the geometric data. Their agreement is not automatic: a hidden extension may remain observationally silent without being generated by the initially declared groupoid.

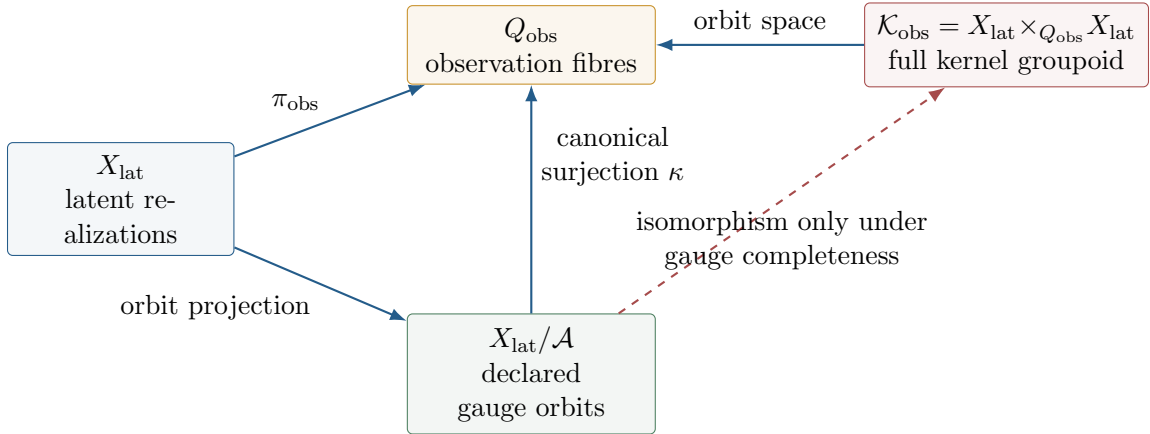


Figure 1: Top-down and bottom-up constructions meet at the kernel groupoid. The declared groupoid embeds into the observational kernel under gauge soundness; it has the same orbit space only under gauge completeness.

2 Observation quotient and kernel groupoid

Definition 2.1 (Observation law and quotient). Let $\mathfrak{F} : X_{\text{lat}} \rightarrow X_{\text{obs}}$ be the declared observation law. Define

$$x \sim_{\text{obs}} y \iff \mathfrak{F}(x) = \mathfrak{F}(y), \quad Q_{\text{obs}} = X_{\text{lat}} / \sim_{\text{obs}}.$$

Write $\pi_{\text{obs}} : X_{\text{lat}} \rightarrow Q_{\text{obs}}$ for the quotient map.

Definition 2.2 (Observational kernel groupoid). The kernel groupoid of π_{obs} is

$$\mathcal{K}_{\text{obs}} = X_{\text{lat}} \times_{Q_{\text{obs}}} X_{\text{lat}} = \{(x, y) : \pi_{\text{obs}}(x) = \pi_{\text{obs}}(y)\} \rightrightarrows X_{\text{lat}},$$

with source $s(x, y) = x$, target $t(x, y) = y$, inverse $(x, y)^{-1} = (y, x)$, and composition $(y, z) \circ (x, y) = (x, z)$.

Proposition 2.3 (Canonical quotient). *The orbits of $\mathcal{K}_{\text{obs}}$ are exactly the observation fibres, hence*

$$X_{\text{lat}}/\mathcal{K}_{\text{obs}} \cong Q_{\text{obs}}.$$

Proof. An element y lies in the $\mathcal{K}_{\text{obs}}$ -orbit of x precisely when $(x, y) \in \mathcal{K}_{\text{obs}}$, which is precisely $\pi_{\text{obs}}(x) = \pi_{\text{obs}}(y)$. $\square$

Definition 2.4 (Gauge soundness and completeness). A declared automorphism groupoid $\mathcal{A} \rightrightarrows X_{\text{lat}}$ is *sound* when all of its arrows lie in $\mathcal{K}_{\text{obs}}$. It is *orbit complete* when

$$\mathcal{A} \cdot x = [x]_{\text{obs}} \quad \text{for every } x \in X_{\text{lat}}.$$

Theorem 2.5 (Orbit-quotient comparison). *Gauge soundness induces a canonical surjection*

$$\kappa : X_{\text{lat}}/\mathcal{A} \longrightarrow Q_{\text{obs}}, \quad [x]_{\mathcal{A}} \longmapsto [x]_{\text{obs}}.$$

It is bijective exactly when $\mathcal{A}$ is orbit complete.

Proof. Soundness makes κ well defined. Surjectivity follows from surjectivity of π_{obs} . Injectivity means that any two points in one observation fibre belong to one $\mathcal{A}$ -orbit, which is exactly orbit completeness. $\square$

Remark 2.6 (No converse from invisibility alone). The implication

$$y = a \cdot x \implies x \sim_{\text{obs}} y$$

is gauge soundness. Its converse is an additional theorem target. Silent extensions are direct counterexamples when the declared groupoid does not move the hidden extension coordinates.

3 Saturation is not gauge completeness

Let $I = (I_1, \dots, I_m)$ be a family of $\mathcal{A}$ -invariant observable functions. There are three logically distinct closure properties.

Definition 3.1 (Three saturation locks). On a regular stratum $S \subset X_{\text{lat}}$:

1. *Differential saturation*: the descended map $\bar{I} : S/\mathcal{A} \rightarrow \mathbb{R}^m$ has full differential rank.
2. *Global orbit separation*: $I(x) = I(y)$ implies $y \in \mathcal{A} \cdot x$ for $x, y \in S$, including discrete residuals.
3. *Gauge completeness*: $\mathcal{A} \cdot x = [x]_{\text{obs}}$.

Proposition 3.2 (Strict hierarchy). *None of these three properties collapses automatically into the next. Full local rank need not remove a discrete sign ambiguity; globally separating $\mathcal{A}$ -orbits need not identify silent extensions outside $\mathcal{A}$; and gauge completeness does not reconstruct latent coordinates inside an observation fibre.*

Proof. The map $(u, v) \mapsto (u^2, v)$ has generic full rank but identifies (u, v) and $(-u, v)$. A hidden coordinate h absent from $\mathfrak{F}$ produces several declared orbits in one observation fibre unless h -motions are included. Finally, an orbit quotient records the orbit, not a preferred representative inside it. $\square$

The saturated second-order RG fibre therefore coordinates a generic orbit space relative to its declared observation law. The resolved eigenvalues, null eigenvalues, canonical correlations, and mixed traces close continuous magnitude and orientation freedom on the regular stratum; rank walls and isotropy jumps remain stratified loci rather than failures of the construction.

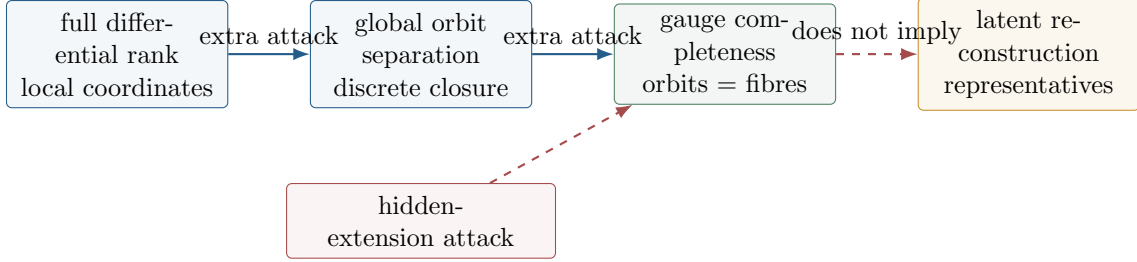


Figure 2: The saturation ladder. Local rank, global orbit separation, and gauge completeness are separate obligations. None reconstructs a preferred latent representative.

4 The mutual fibre and horizontal observable structure

Assume temporarily that π_{obs} is differentiable on a regular stratum. Set

$$V_x^{\text{obs}} = \ker D\pi_{\text{obs}}|_x, \quad V_x^{\mathcal{A}} = T_x(\mathcal{A} \cdot x).$$

Gauge soundness gives

$$V_x^{\mathcal{A}} \subseteq V_x^{\text{obs}}.$$

The residual silent quotient

$$S_x^{\text{silent}} = V_x^{\text{obs}} / V_x^{\mathcal{A}}$$

measures infinitesimal observation-fibre directions not generated by the declared groupoid.

Theorem 4.1 (Regular observable tangent sequence). *If π_{obs} is a submersion on a regular stratum, then*

$$0 \longrightarrow V^{\text{obs}} \longrightarrow TX_{\text{lat}} \xrightarrow{D\pi_{\text{obs}}} \pi_{\text{obs}}^* TQ_{\text{obs}} \longrightarrow 0$$

is exact. Consequently,

$$T_{[x]}Q_{\text{obs}} \cong T_x X_{\text{lat}} / V_x^{\text{obs}}.$$

Under infinitesimal gauge completeness, $V_x^{\mathcal{A}} = V_x^{\text{obs}}$ and the right-hand side is the usual regular orbit tangent quotient.

Proof. This is the standard tangent sequence of a submersion. The final identification follows upon equality of the two vertical spaces. $\square$

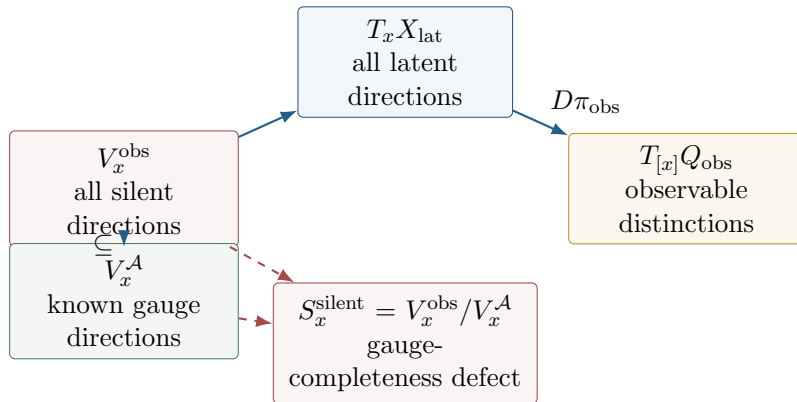


Figure 3: The mutual fibre structure. The quotient removes the full observation vertical space; the declared groupoid may explain only part of it.

At singular points the ranks of isotropy and orbit directions jump. The object is then not one vector bundle quotient. Proper Lie groupoid orbit spaces carry natural orbit-type stratifications and slice models; this is precisely the mathematical setting needed for RG’s Weyl-chamber walls, PSD-cone faces, and rank-changing fibres [1, 2].

5 Descent of observable transport

Definition 5.1 (Projectable transport). A latent vector field W is observationally projectable when there is a field W_{obs} on the regular quotient satisfying

$$D\pi_{\text{obs}} \circ W = W_{\text{obs}} \circ \pi_{\text{obs}}.$$

Equivalently, $D\pi_{\text{obs}}(W_x)$ is independent of the representative x in an observation fibre.

Theorem 5.2 (Unique transport descent). *A projectable latent transport has a unique descended transport W_{obs} . A field whose observable component depends on a gauge or other silent coordinate cannot descend.*

Proof. Define $W_{\text{obs}}([x]) = D\pi_{\text{obs}}(W_x)$. Projectability gives well-definedness and surjectivity of π_{obs} gives uniqueness. If two representatives give different projections, no single value can be assigned to the quotient class. $\square$

This identifies the role of W_{obs} : it is neither the quotient nor the groupoid. It is the transport that survives both. Vertical terms may change latent representatives, including gauge frame transport and silent hidden motion, while the projected horizontal term is the observable continuation law.

6 Descent of the admissible cone

Let $\mathcal{C}_{\text{adm}x} \subset T_x X_{\text{lat}}$ be the latent set of admissible instantaneous transport directions.

Definition 6.1 (Cone projectability). The family $\mathcal{C}_{\text{adm}}$ is observationally projectable when

$$D\pi_{\text{obs}}(\mathcal{C}_{\text{adm}x}) = D\pi_{\text{obs}}(\mathcal{C}_{\text{adm}y}) \quad \text{whenever } x \sim_{\text{obs}} y,$$

with the appropriate tangent identifications on the regular stratum.

Theorem 6.2 (Observable cone descent). *If $\mathcal{C}_{\text{adm}}$ is cone-projectable, then*

$$\mathcal{C}_{\text{obs}[x]} := D\pi_{\text{obs}}(\mathcal{C}_{\text{adm}x}) \subset T_{[x]} Q_{\text{obs}}$$

is well defined. Every projectable admissible transport obeys

$$W_{\text{obs}}([x]) \in \mathcal{C}_{\text{obs}[x]}.$$

Conversely, hidden-dependent variation of the projected cone opening obstructs descent.

Proof. Representative independence is exactly cone projectability. The transport claim follows by applying $D\pi_{\text{obs}}$ to $W_x \in \mathcal{C}_{\text{adm}x}$. The converse is immediate from failure of representative independence. $\square$

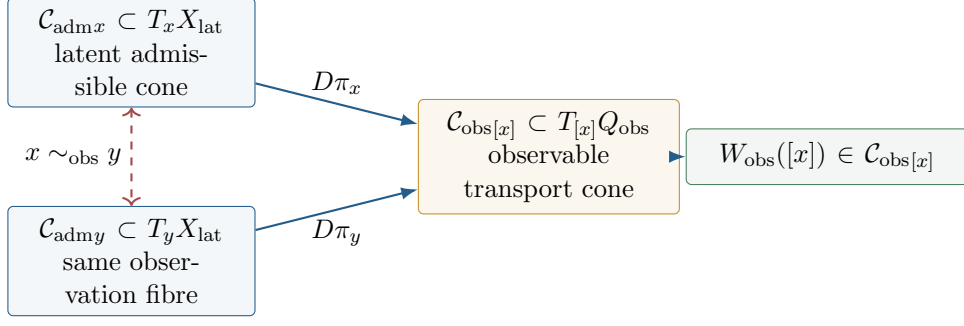


Figure 4: Cone projectability is stronger than gauge invariance alone: the projection must agree across the entire observation fibre, including silent extensions not initially generated by gauge.

7 From the observable cone to conformal geometry

A cone need not be quadratic. Therefore the bridge to a Lorentzian conformal metric requires an explicit gate.

Assumption 7.1 (Regular quadratic cone gate). On a regular quotient stratum of dimension $n \geq 3$, each $\mathcal{C}_{\text{obs}q}$ is the future causal cone of a nondegenerate Lorentzian quadratic form, varies with the required regularity, and has a fixed time orientation.

Theorem 7.2 (Pointwise conformal reconstruction). *Under the regular quadratic cone gate, the null boundary of $\mathcal{C}_{\text{obs}q}$ determines a unique positive metric ray*

$$[g_q] = \{\Omega(q)^2 g_q : \Omega(q) > 0\}.$$

Thus the cone field determines a Lorentzian conformal class $[g]$, not a metric representative.

Proof. Two nondegenerate Lorentzian quadratic forms with the same null cone in dimension at least three are proportional. The declared timelike sign and time orientation select the positive proportionality ray. Regular variation gives a conformal field. $\square$

At the global level, established causal reconstruction results similarly show, under distinguishing causality and Lorentzian realizability assumptions, that causal structure determines conformal spacetime geometry. They do not apply to an arbitrary preorder and do not fix the conformal factor [3, 4].

Proposition 7.3 (Nonquadratic branch). *If the descended admissible cone is regular and homogeneous but not the zero set of a nondegenerate quadratic form, the conclusion $\mathcal{C}_{\text{obs}} \Rightarrow [g]$ fails. The appropriate realization may instead be Lorentz–Finsler, dispersive, multicone, or more general characteristic geometry.*

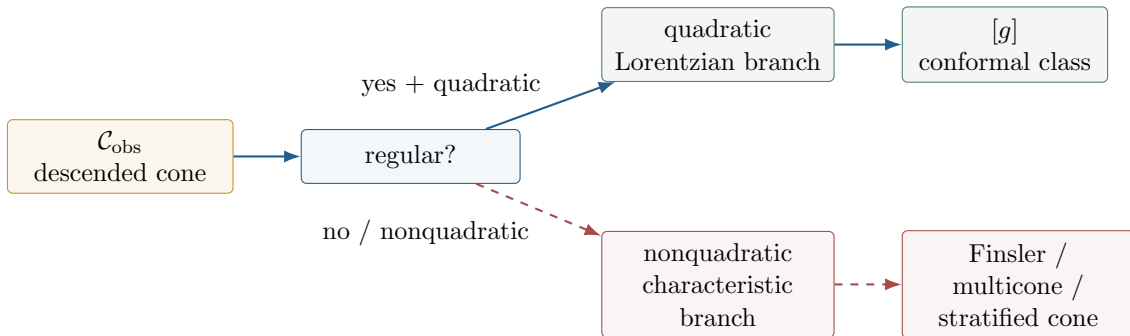


Figure 5: Geometric realization is a branch, not an automatic consequence of quotienting.

8 Scale selection and strict transport

The conformal class retains a positive function $\Omega(q)$. A single clock worldline cannot determine that function away from the worldline.

Definition 8.1 (Strict scale transport). A representative-preserving transport P_{qr} is strict when

$$P_{qr}^* g_r = g_q.$$

It is conformal or Weyl-compatible when only $P_{qr}^* g_r = \rho_{qr}^2 g_q$ is required.

Theorem 8.2 (Global scale closure). *Suppose two representatives g and $\tilde{g} = \Omega^2 g$ are compatible with the same strict transport on a transport-connected quotient. Then Ω is constant on that connected component. One physical scale anchor fixes the remaining constant. If edgewise Weyl compensation is admissible, nonconstant Ω remains possible.*

Proof. Strict compatibility for both representatives gives $\Omega(r)^2 g_q = \Omega(q)^2 g_q$, and nondegeneracy plus positivity yields $\Omega(r) = \Omega(q)$. Connectedness propagates equality. Under Weyl transport, the ratio can be absorbed into ρ_{qr} . $\square$

An operational coherence duration $\mathcal{T} = -\log C$ supplies candidate local scale data when coherence is multiplicative. Global metric selection requires these local calibrations to be integrable: overlap differences must form an exact cocycle, with zero accumulated scale holonomy around closed transport loops.

9 The Observable Orbit–Cone Theorem

Theorem 9.1 (Observable Orbit–Cone Theorem). *Let $\pi_{\text{obs}} : X_{\text{lat}} \rightarrow Q_{\text{obs}}$ be an observational quotient, let $\mathcal{A} \rightrightarrows X_{\text{lat}}$ be a declared automorphism groupoid, let I be a family of declared observable invariants, let W be a latent transport, and let $\mathcal{C}_{\text{adm}} \subset TX_{\text{lat}}$ be an admissible latent cone family. On a regular stratum, assume:*

1. $\mathcal{A}$ is gauge sound;
2. if orbit-space equality is claimed, $\mathcal{A}$ is orbit complete;
3. I is locally saturated and globally separates the declared regular orbits, including discrete residuals;
4. W is observationally projectable;
5. $\mathcal{C}_{\text{adm}}$ is observationally cone-projectable;
6. the descended cone is regular and quadratic if Lorentzian conformal reconstruction is claimed.

Then:

1. there is a canonical surjection $X_{\text{lat}}/\mathcal{A} \rightarrow Q_{\text{obs}}$, which is an isomorphism exactly under gauge completeness;
2. saturated invariants provide regular coordinates on the completed orbit space but do not reconstruct latent representatives;
3. W descends uniquely to W_{obs} ;
4. $\mathcal{C}_{\text{adm}}$ descends to $\mathcal{C}_{\text{obs}} \subset TQ_{\text{obs}}$ and $W_{\text{obs}}(q) \in \mathcal{C}_{\text{obs}q}$;
5. a regular quadratic $\mathcal{C}_{\text{obs}}$ determines a conformal class $[g]$;

6. an independent integrable scale calibration is required to select g .

At singular orbit-type or rank-changing strata, ordinary tangent quotients are replaced by slices, strata, and tangent cones.

Proof. The first statement is Theorem 2.5; the second is the saturation hierarchy; the third is Theorem 5.2; the fourth is Theorem 6.2; the fifth is pointwise conformal reconstruction; and the sixth is the scale-closure theorem. Orbit-dimension jumps invalidate a single smooth quotient tangent bundle and require stratified slice models. $\square$

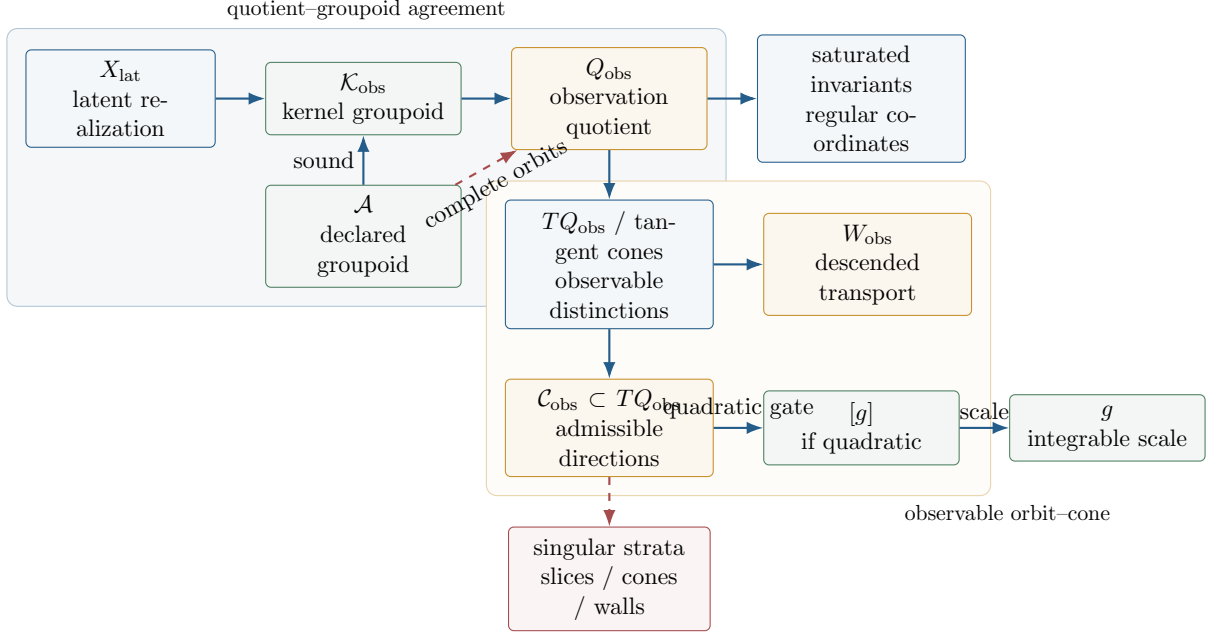


Figure 6: The full RG architecture. Each arrow has a separate gate; no layer is asked to perform the logical work of another.

10 Why this is the full geometry of RG

The theorem reconciles the descriptions already present in RG.

Observable reconstruction. The data (J, Σ, g_O) determine (R, N, Σ) equivariantly. Thus one RG geometry is a canonical observable object up to admissible representation gauge. The full covariance is required; Fisher information alone is a many-to-one Schur-complement summary and cannot reconstruct the null block.

Stratified fibre. At fixed observable base point, the quotient by the frame gauge produces closed eigenvalue chambers, principal-angle data, mixed orientation invariants, PSD cones, and the Schur floor. Colliding eigenvalues and rank loss are orbit-type walls, not removable coordinate defects. Proper Lie groupoid orbit spaces are naturally stratified and admit slice descriptions, length-space structures, and basic differential forms [1].

Horizontal transport. Paths, holonomy, monodromy, and continuation live in projectable horizontal directions. They descend as W_{obs} only after all observation-vertical variation has been removed. This includes both declared gauge directions and any additional silent-extension directions.

Observable cone. Admissibility selects a cone in the horizontal quotient. Its quadratic regular branch determines conformal causal geometry; its singular and nonquadratic branches retain the broader stratified character already visible in the RG fibre.

Scale and physical realization. The cone fixes causal directions, not metric units. An integrable operational clock/ruler calibration selects a representative. Identification with a physical light cone and use of GR dynamics are later realization steps, not premises of the quotient theorem.

Table 1: One RG structure, with each layer assigned one logical job.

Layer	Determines	Does not determine
Observation quotient	indistinguishable classes	experimental mechanism generating every fibre
Declared groupoid	known gauge motions and orbit structure	all silent extensions without completeness
Saturated invariants	coordinates separating regular completed orbits	latent representatives or singular smoothing
Vertical kernel	all infinitesimally silent directions	a preferred horizontal lift
W_{obs}	descended observable continuation	arbitrary hidden-dependent dynamics
$\mathcal{C}_{\text{obs}}$	admissible observable direction field	quadraticity or absolute scale
Quadratic cone	conformal Lorentzian class $[g]$	metric representative g
Scale cocycle	calibrated representative when integrable	Einstein dynamics

11 Attack battery and reproducibility

The final battery is synthetic and adversarial. Its role is to verify finite models of each logical gate and to seek counterexamples to collapsed claims. It combines the original 20-check orbit-cone battery with six final closure attacks. Seed 20260824 passed all

26/26

registered checks.

Table 2: Selected hostile outcomes.

Attack	Recorded result	Consequence
Declared gauge vs observation fibres	15 orbits, 5 fibres	automatic gauge completeness rejected
Completed groupoid	5 orbits, 5 fibres	orbit-space equality recovered
Gauge-completeness gap	10	silent extension classes quantified
Basic transport projection error	0	W_{obs} descends
Nonbasic hidden dependence	0.4 gap	representative-dependent transport rejected
Quadratic cone reconstruction	1.62×10^{-17} ray error	conformal metric ray recovered
Nonquadratic cone attack	4.95×10^{-2} singular ratio	quadraticity gate required
Strict transport scale nullity	1	one connected global scale mode
Full-fibre cone projectability	0 projection error	observable cone well defined
Hidden-dependent cone opening	positive gap	nonprojectable cone rejected
Isotropy jump	rank change 1	stratified tangent cone required
Scale-loop obstruction	nonzero residual	global calibration requires integrability

The artifacts are:

- `observable_orbit_cone_attack.py`;
- `observable_orbit_cone_attack.json`;
- `observable_orbit_cone_final_attack.py`;
- `observable_orbit_cone_final_attack.json`.

All numerical results are reproducibility metadata. They are not premises of the analytic theorem ladder.

12 Non-claims

Non-claim 12.1. This paper does not claim that an initially declared automorphism groupoid is automatically gauge complete, that full differential rank implies global invariant separation, or that saturation reconstructs latent ontology.

Non-claim 12.2. The ordinary tangent quotient is asserted only on regular strata. Singular rank and isotropy loci require orbit-type stratification, slices, and tangent cones. No smooth global manifold is claimed for the complete RG object.

Non-claim 12.3. A cone is not automatically Lorentzian quadratic. Conformal reconstruction is conditional on the quadratic gate, and selection of a metric representative is conditional on independent integrable scale calibration.

Non-claim 12.4. No Einstein field equation, stress tensor, quantum measurement law, speed of light, or physical spacetime ontology is derived. Those are possible physical realizations and calibrations of the observational geometry, not consequences of quotient universality alone.

13 Conclusion

The quotient and automorphism-groupoid pictures agree through a precise comparison, not through a slogan. The observational kernel groupoid contains all invisible pairs. A declared automorphism groupoid explains a subfamily of those invisible motions; it reproduces the top-down quotient exactly only when its orbits are complete. Saturated invariants then coordinate the regular completed orbit space, while rank and isotropy changes retain the singular fibre geometry.

The common differential structure is

$$T_x X_{\text{lat}} / V_x^{\text{obs}} \cong T_{[x]} Q_{\text{obs}}$$

on regular strata, with slices and tangent cones replacing it at singular points. Projectable transport descends uniquely to W_{obs} . Projectable admissibility descends to $\mathcal{C}_{\text{obs}}$. A regular quadratic $\mathcal{C}_{\text{obs}}$ determines $[g]$, and an integrable scale field selects g .

Top-down quotient and bottom-up groupoid meet in the observational kernel.

Their mutual geometry is the stratified observable horizontal tangent–cone structure.

W_{obs} is its projectable transport; $\mathcal{C}_{\text{obs}}$ is its admissible cone; a quadratic regular cone determines conformal geometry.

References

- [1] M. J. Pflaum, H. Posthuma, and X. Tang, “Geometry of orbit spaces of proper Lie groupoids,” *Journal für die reine und angewandte Mathematik* 694 (2014), 49–84.

- [2] N. T. Zung, “Proper groupoids and momentum maps: linearization, affinity, and convexity,” *Annales scientifiques de l’École Normale Supérieure* 39 (2006), 841–869.
- [3] S. W. Hawking, A. R. King, and P. J. McCarthy, “A new topology for curved space–time which incorporates the causal, differential, and conformal structures,” *Journal of Mathematical Physics* 17 (1976), 174–181.
- [4] D. B. Malament, “The class of continuous timelike curves determines the topology of space–time,” *Journal of Mathematical Physics* 18 (1977), 1399–1404.
- [5] M. Goresky and R. MacPherson, *Stratified Morse Theory*, Springer, 1988.
- [6] D. Luna, “Slices étales,” *Bulletin de la Société Mathématique de France, Mémoire* 33 (1973), 81–105.
- [7] S. Amari and H. Nagaoka, *Methods of Information Geometry*, AMS/Oxford, 2000.
- [8] F. Zhang (ed.), *The Schur Complement and Its Applications*, Springer, 2005.